



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-2027
COURSE HANDOUT (PART II)

Date: 03/08/2026

In addition to Part-I (general handout for all courses appended to this time table) this portion gives further details pertaining to the course.

Course No.: CS F415

Course Title: **Data Mining**

Instructor-in-charge: **YASHVARDHAN SHARMA** (yash@pilani.bits-pilani.ac.in)

Lab Instructor : Anubhav Pandey (p20220020@pilani.bits-pilani.ac.in)

1. Course Description :

Introduction to Data Mining, fundamental concepts; motivation and applications; role of data warehousing in data mining; challenges and issues in data mining; Knowledge Discovery in Databases (KDD); role of data mining in KDD; algorithms for data mining; tasks like decision-tree construction, finding association rules, sequencing, classification, and clustering; applications of neural networks and machine learning for tasks of classification and clustering.

2. Objective and Scope

The course explores the concepts and techniques of data mining, a promising and flourishing frontier in database systems. Data Mining is automated extraction of patterns representing knowledge implicitly stored in large databases, data warehouses, and other massive information repositories. It is a decision support tool that addresses unique decision support problems that cannot be solved by other data analysis tools such as Online Analytical Processing (OLAP). The course covers data mining tasks like constructing decision trees, finding association rules, classification, and clustering. The course is designed to provide students with a broad understanding in the design and use of data mining algorithms. The course also aims at providing a holistic view of data mining. It will have database, statistical, algorithmic and application perspectives of data mining.

3. Text Book

- i) Tan P. N., Steinbach M & Kumar V. *“Introduction to Data Mining”* Pearson Education, 2016.

4. Reference Books

- i) Han J & Kamber M, *“Data Mining: Concepts and Techniques”*, Morgan Kaufmann Publishers, Third Edition, 2012
ii) Zaki MJ & Wagner M JR, *“Data Mining and Analysis-Fundamental Concepts and Algorithms”* Cambridge Univ Press, 2014.

4. Course Plan

Lecture No.	Learning Objective	Topic(s)	Chapter Reference
1-2	To understand the definition and applications of Data Mining	Introduction to Data Mining • Motivation • What is Data Mining? • Data Mining Tasks • Issues in Data Mining • Applications	1+Class Notes
3-5	To understand types of data and to improve the quality of data and efficiency and the ease of	Data Preprocessing • Types of data • Data Quality • Data preprocessing	2



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	the mining process.	Similarity and Dissimilarity Measures	
6	To study how to investigate the data	Data Exploration - Data Set & its Statistics - Visualization - OLAP & Multidimensional Data Analysis	3 Self Study
7-10	To understand Classification and its applications	Classification - Introduction - Applications - Decision Tree based Algorithms - Model Over-fitting - Performance Evaluation of a Classifier - Comparing Classifiers	4+Class Notes
11-15	To study the alternative approaches for Classification	Classification: Alternative Techniques - Rule Based Classifier - Nearest Neighbor Classifier - Bayesian Classification - Support Vector Machine - Ensemble Classifiers - Class Imbalance Problem - Multiclass Problem	5
16-19	To understand applications of Association Rule Mining and algorithms to find them	Association Rule Mining - Introduction - Applications - Market-Basket Analysis - Frequent Itemsets - Apriori Algorithm - Alternative Methods	6
20-23	To understand methods and need for finding complex Association Rules	Advanced Association Rule Mining - Generalized Association Rules - Multilevel Association Rules - Multidimensional Association Rules - Temporal Association Rules - Infrequent Patterns - Constrained Based Association Rules	7+Class Notes
24-28	To understand applications and algorithms for Clustering	Clustering - Introduction - Applications - Partitioning Algorithms - Hierarchical Algorithms - Density based Algorithms - Cluster Evaluation	8
29-33	To study advanced topics in cluster analysis	Clustering: Additional Issues and Algorithms - Characteristics of Data, Clusters and clustering Algorithms - Graph Based Clustering - Scalable Clustering Algorithms	9
34-35	To understand detection of anomalies & their causes	Anomaly Detection - Preliminaries - Statistical Approaches - Proximity based Outlier Detection - Density based Outlier Detection	10



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		Clustering Based Techniques	
36-40	To introduce advanced topics in Data Mining	Advanced Topics - Web Mining - Incremental Algorithms for Data Mining - Stream Data Mining	Class Notes

5. Evaluation Schedule

Component	Duration	Weightage(%)	Date & Time	Venue	Remarks
Mid Sem Exam	90 Mins.	25	09/10AN1		Closed Book
Quiz/Labs/OnlineTest/Project		35	To be announced		
Comprehensive	3 Hours	40	14/12 FN		Partly open

7. Labs

Two hour lab will be conducted every week. Students will be applying the concepts of data mining on the problems Python Programming and IBM SPSS Modeler. Students will also be exposed to modeling of the problems.

8. Project

Project/Assignment(s) (programming/reading) will be given to the students. This will immensely help the students in gaining a better understanding of the subject.

9. Chamber Consultation Hours

Tuesday 4pm to 5pm @6120-A

10. Make-up Policy: Prior Permission is must and Make-up shall be granted only in genuine cases based on individual's need and circumstances.

11. Notices

All the notices concerning this course will be displayed on the **CSIS Notice Board** or **course website** (<https://nalanda.bits-pilani.ac.in/>)

Instructor-in-charge



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